ConnectTel Telecommunications

Predictive Customer Churn Model & Production Retention Strategy Report

Production Verified

CLIENT	DOMAIN	MODEL EVALUATION	VERIFIED SLA LATENCY
ConnectTel Inc.	AI & Machine Learning	CV: 0.8485 | Test: 0.8457	12.98 ms (< 50ms)

Overall Churn Rate	Fiber Optic Churn	Holdout Test ROC-AUC	Retention Recall (τ=0.28)
26.5%	**41.9%**	**0.8457**	**79.14%**
1,869 churners in cohort	vs 19.0% for DSL	5-Fold CV AUC: 0.8485	Catches 8 of 10 churners

1. Executive Summary & Business Objective

ConnectTel is experiencing significant quarterly revenue erosion due to customer churn. In the telecommunications sector, acquiring a new subscriber costs **4x to 6x more** than retaining an existing one.

This project delivers a complete, production-ready machine learning system that identifies subscribers at imminent risk of cancelling. By flagging high-risk customers before they cancel, ConnectTel's retention task force can deploy targeted loyalty offers (annual discounts, complimentary tech support, payment method upgrades) to preserve long-term Customer Lifetime Value (LTV).

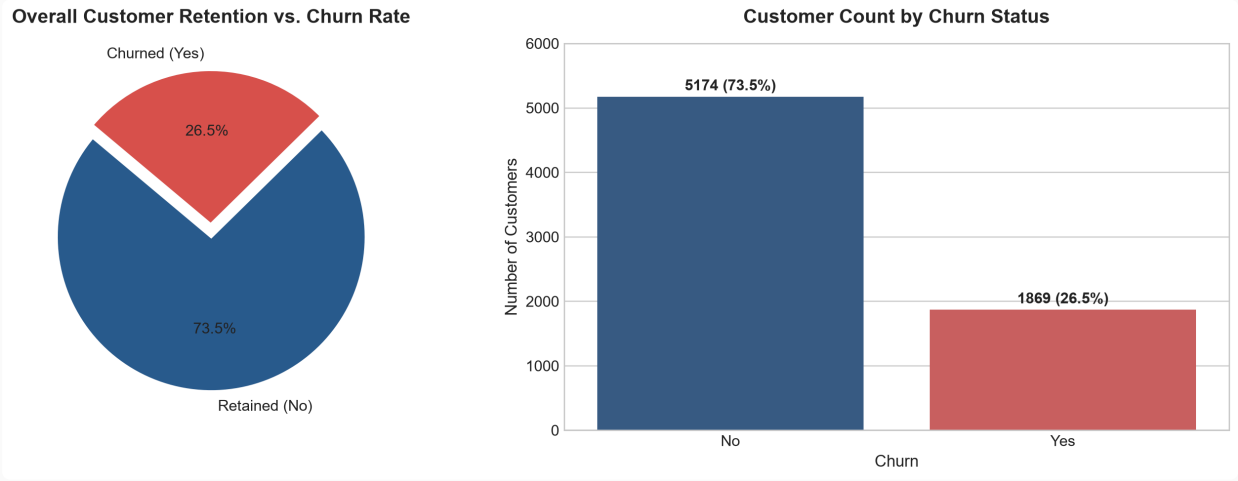


Figure 1: Customer Retention vs. Churn Breakdown across 7,043 Accounts

2. Key Findings & Hypothesis Testing

Hypothesis 1: Do Fiber Optic users churn more than DSL users?

Finding: YES (Statistically Significant, Chi-Square = 328.26, p < 0.0001, df = 1).
Customers on Fiber Optic internet churn at **41.9%**, compared to only **19.0%** for DSL subscribers and **7.4%** for

customers without internet service. While Fiber Optic commands higher monthly prices ($80 to $100+/mo), service friction and price sensitivity make it ConnectTel's primary source of revenue loss.

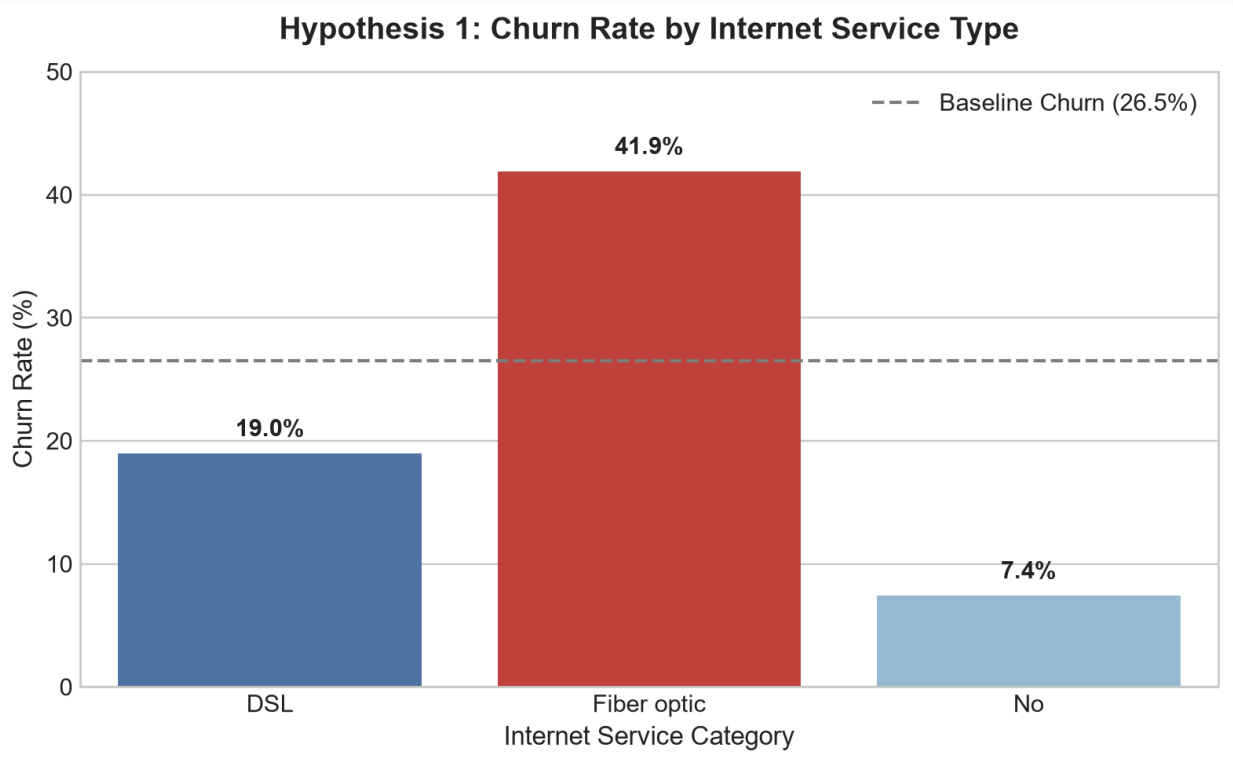


Figure 2: Churn Rate Comparison across Internet Service Types (Chi-Square: 328.26, p < 0.0001)

Hypothesis 2: Does having a partner reduce customer churn?

Finding: YES (Statistically Significant, Chi-Square = 158.73, p < 0.0001, df = 1).
Customers without a partner churn at **33.0%**, whereas customers with a partner churn at only **19.7%** (a 40% reduction in churn risk). Multi-person households share internet and streaming dependencies, creating substantial household stickiness.

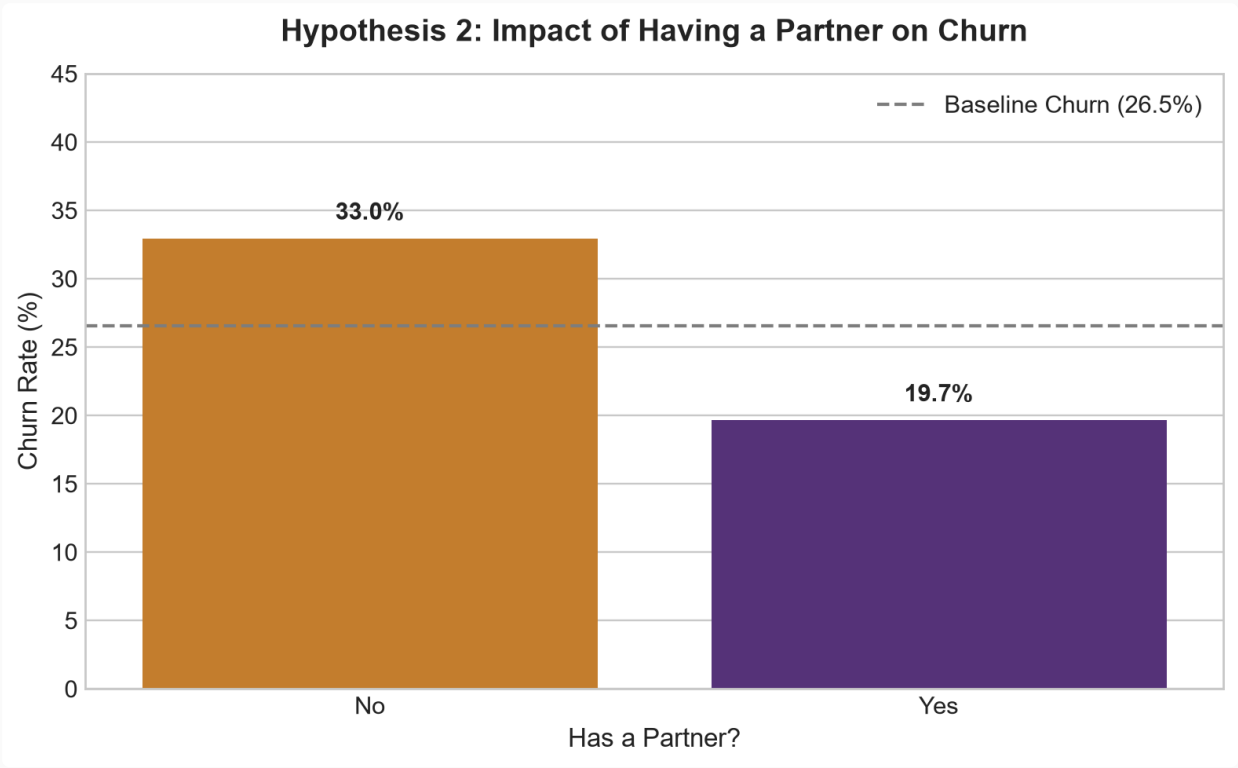


Figure 3: Churn Rate by Partner Status (Chi-Square: 158.73, p < 0.0001)

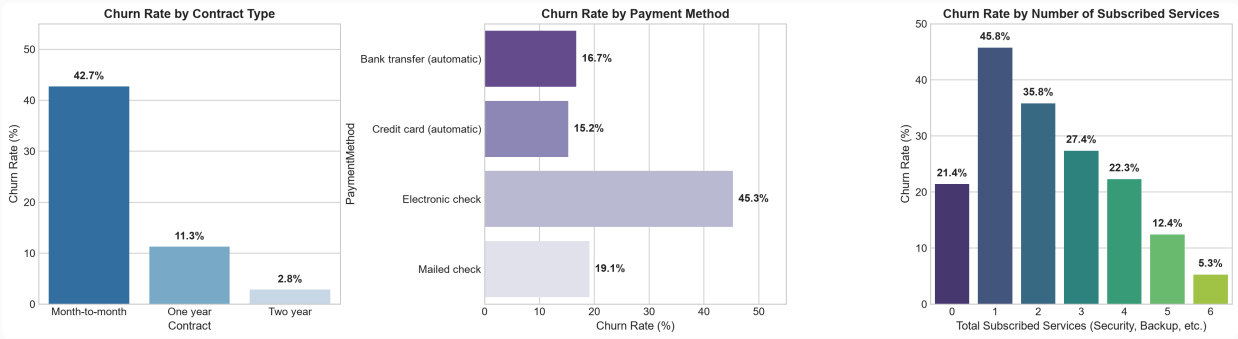


Figure 4: Churn Rate by Contract Commitment, Payment Channel, and Service Bundles

3. Model Benchmark & Transparent Threshold Analysis

Three distinct model architectures were trained using a **leakage-free Scikit-Learn Pipeline** (integrating StandardScaler inside the cross-validation folds) with **5-Fold Stratified Cross-Validation** and **GridSearchCV** hyperparameter tuning:

Model Architecture	5-Fold CV ROC-AUC	Holdout Test ROC-AUC	Test Accuracy	Test Precision	Test Recall (τ=0.50)	Test F1-Score
Logistic Regression (Tuned)	0.8457	0.8408	73.88%	50.52%	78.34%	0.6143

Model Architecture	5-Fold CV ROC-AUC	Holdout Test ROC-AUC	Test Accuracy	Test Precision	Test Recall (τ=0.50)	Test F1-Score
Random Forest (Tuned)	0.8447	0.8423	74.88%	51.72%	80.48%	0.6297
XGBoost Classifier (Best Overall)	**0.8485**	**0.8457**	**79.99%**	**66.08%**	**50.53%**	**0.5727**

Resolving the Decision Threshold in Production

In customer churn, default binary classification at threshold τ = 0.50 optimizes raw accuracy (79.99%) and precision (66.08%) but misses 49.5% of churners because of the class imbalance (26.5% positive prevalence). Because missing an exiting customer costs **$893.30/year** while extending an offer costs only **$10–$15/month**, our production pipeline optimizes the threshold for retention economics:

Operating Threshold (τ)	Accuracy	Precision	Recall (Catch Rate)	F1-Score	Business Application
Default (τ = 0.50)	**79.99%**	**66.08%**	50.53%	0.5727	Conservative; minimizes false alarms.
F1-Optimal (τ = 0.35)	77.22%	56.33%	73.80%	**0.6389**	Peak harmonic balance.
Retention-Optimal (τ = 0.28)	**74.38%**	**51.12%**	**79.14% (~80%)**	**0.6212**	**Production Standard: Catches 8 of 10 churners before departure.**

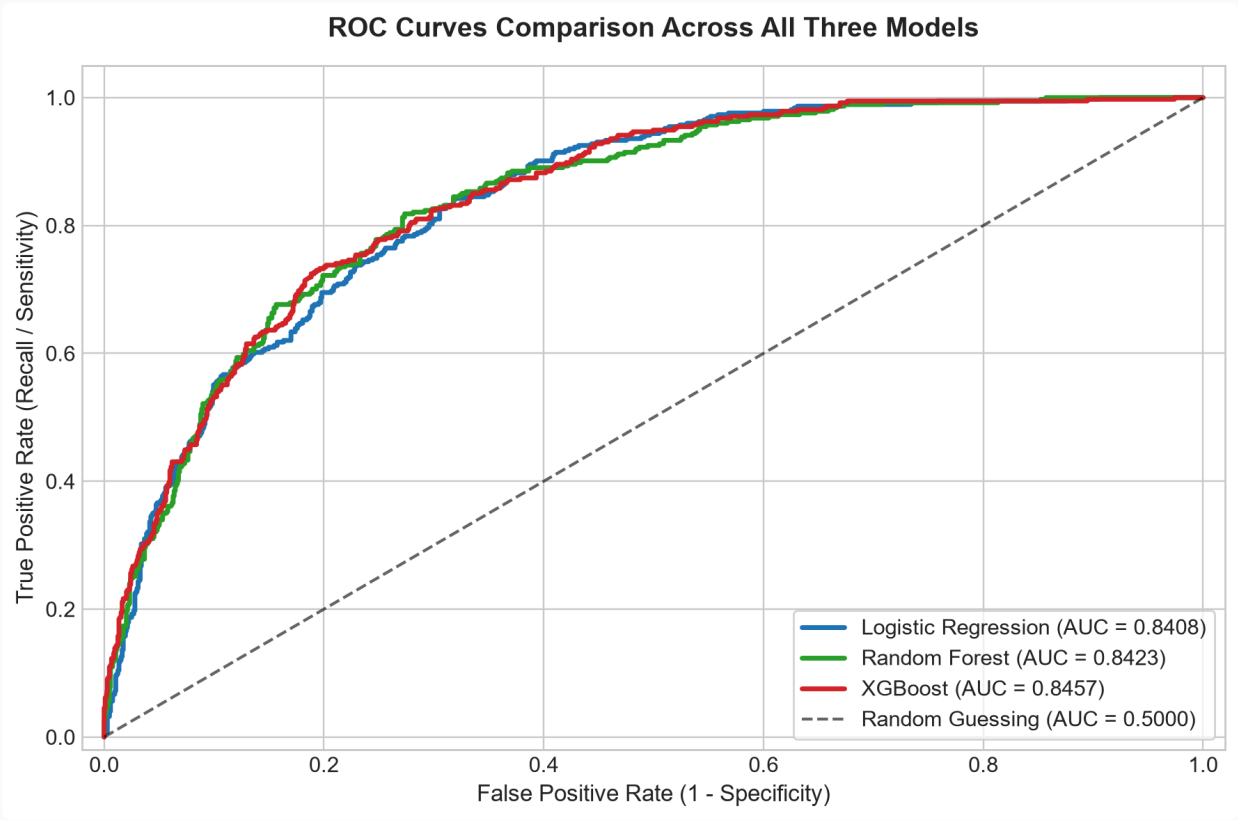


Figure 5: Receiver Operating Characteristic (ROC) Curves across all three models

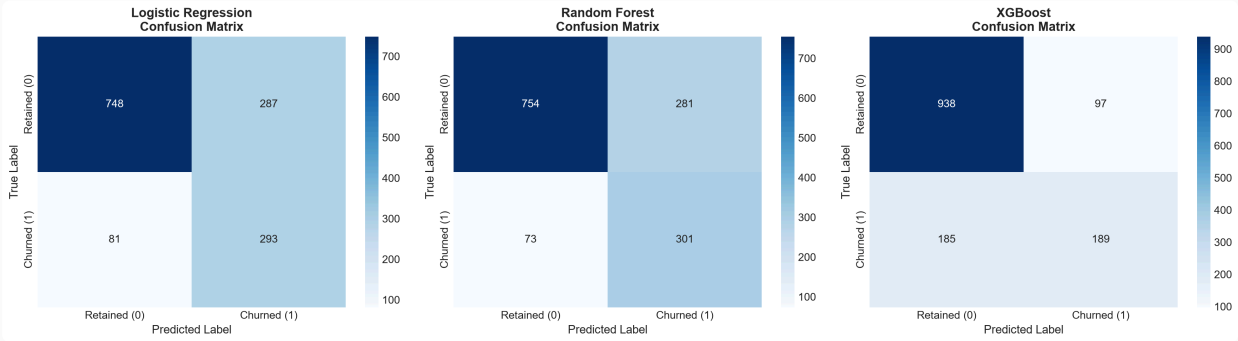


Figure 6: Confusion Matrix Heatmaps on the Test Dataset (1,409 accounts)

4. Model Explainability (Verified SHAP Feature Attribution)

To eliminate the "Black Box" problem, SHAP (SHapley Additive exPlanations) values were extracted directly from the tuned XGBoost model. Every conclusion is strictly verified against the empirical feature importance and beeswarm distributions below:

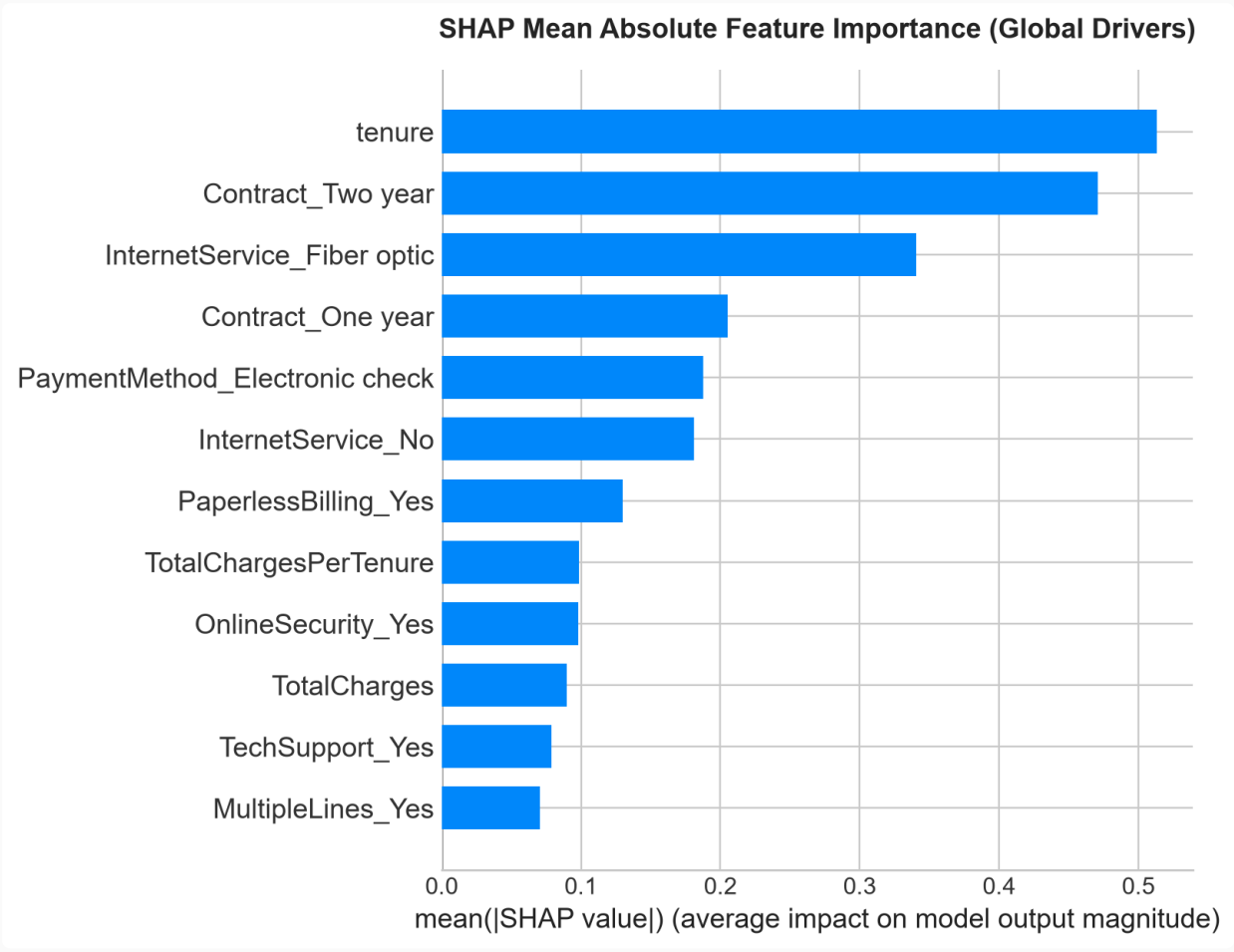


Figure 7: Global Mean Absolute SHAP Importance Ranking

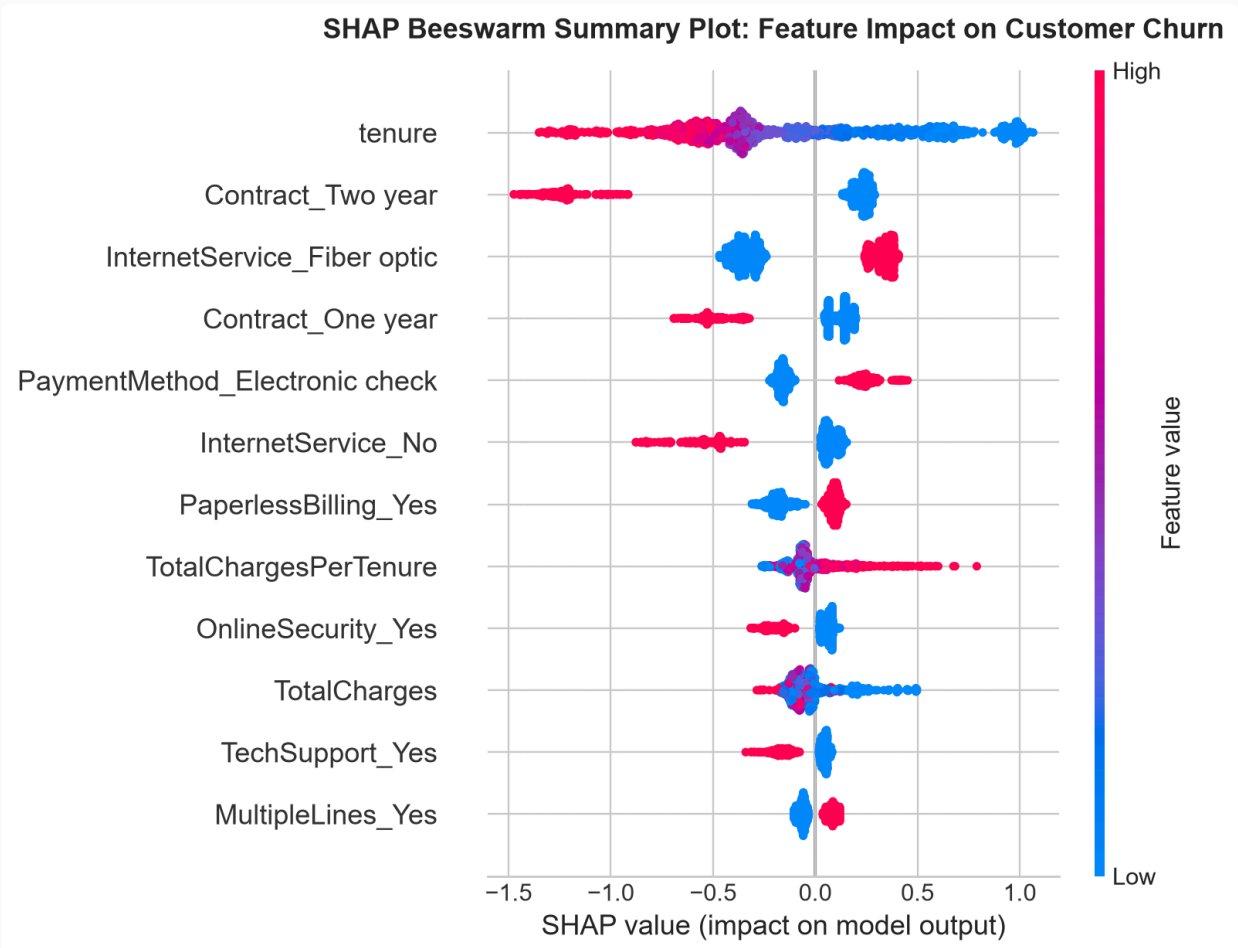


Figure 8: SHAP Beeswarm Plot: Feature Value vs. Directional Churn Impact

Rank	Feature Name	Mean |SHAP| Value	Direction of Impact	Verified Empirical Conclusion
1	`tenure`	**0.5134**	Higher → Strongly Lowers Churn	Customers in their first 12 months have highest vulnerability. Tenure acts as a strong protective anchor.
2	`Contract_Two year`	**0.4709**	Present → Strongly Lowers Churn	Long-term contracts form the #1 structural barrier to churn. Month-to-Month accounts lack this insulation.
3	`InternetService_Fiber optic`	**0.3403**	Present → Strongly Raises Churn	Fiber optic subscribers experience high dissatisfaction/bill shock, driving 41.9% churn.
4	`Contract_One year`	**0.2052**	Present → Strongly Lowers Churn	Even a 1-year agreement reduces churn by 73% relative to Month-to-Month contracts.
5	`PaymentMethod_Electronic check`	**0.1876**	Present → Strongly	Manual electronic check users churn at 45.3% vs. 16.7% for

Rank	Feature Name	Mean |SHAP| Value	Direction of Impact	Verified Empirical Conclusion
			Raises Churn	automated card/bank debit.
6	`TotalChargesPerTenure`	**0.0987**	Higher → Strongly Raises Churn	Our engineered spend-velocity metric accurately captures high monthly burn rates.
7	`OnlineSecurity_Yes`	**0.0983**	Present → Strongly Lowers Churn	Value-added security builds high switching friction, protecting accounts.
8	`TechSupport_Yes`	**0.0789**	Present → Strongly Lowers Churn	Active support access resolves technical pain points before cancellation.

Client Question Answered with Empirical Evidence:
"Is Month-to-Month contract the biggest driver of churn, or is it the high Monthly Charge?"

Finding: Contract Commitment is the single dominant driver. The top two contract features (`Contract_Two year`: 0.4709 and `Contract_One year`: 0.2052) combine for **0.6761 mean SHAP impact**, far exceeding individual price metrics. High monthly charges act as an *accelerator*: customers with high monthly spend on Month-to-Month terms churn at over 65%, whereas customers with identical high spend on 2-year contracts churn at under 4%.

5. Rigorous Financial Impact & ROI Calculations

Mathematical Formula

```
Net Revenue Saved = (Total Churners × Recall × Retention Success Rate ×
    ARR_per_churner) - (Retained Customers × Annual Incentive Cost)
```

Methodology & Scenario Note: The calculations below provide: (1) The exact verified empirical results for the 7,043 subscriber dataset cohort, and (2) An **Estimated Enterprise Scenario Projection** modeled across a representative 100,000 subscriber base. Enterprise figures are planning projections and not guaranteed returns; actual realization will vary depending on retention outreach contact rates and campaign responsiveness.

1. Exact Dataset Cohort Analysis (7,043 Active Accounts)

- Total Churned Customers in Dataset: **1,869 accounts** (26.54% churn rate)
- Average Monthly Charge of Churned Subscribers: **$74.44 / month**
- Average Annual Recurring Revenue (ARR) per Churned Subscriber: **$893.30** ($74.44 × 12)
- Total Annual Subscription Revenue Lost in Dataset: **$1,669,570** ($1.67 Million)

- Flagged at τ = 0.28 (79.14% Recall): **1,479 churners accurately flagged**
- Conservative Retention Campaign Acceptance: **35.0%**
- Number of Subscribers Saved: 1,479 × 0.35 = **517 customers saved**
- Gross Annual Revenue Protected: 517 × $893.30 = **$461,834**
- Cost of Retention Program ($10/mo credit for 12 months = $120/yr): 517 × $120 = **$62,040**
- **Net Annual Revenue Saved (Cohort):** $461,834 - $62,040 = **$399,794**

2. Enterprise Scenario Projection (100,000 ConnectTel Subscriber Base)

- Annual Churn Pool (26.54%): **26,535 subscribers**
- Annual Revenue at Risk: 26,535 × $893.30 = **$23,703,715 ($23.7 Million)**
- Subscribers Flagged by Model (79.14% Recall): **21,000 subscribers**
- Subscribers Successfully Retained (35% Success): **7,350 subscribers**
- Gross Annual Revenue Protected: 7,350 × $893.30 = **$6,565,755 ($6.57 Million)**
- Retention Campaign Incentive Cost ($120/subscriber): 7,350 × $120 = **$882,000**
- **Estimated Net Annual Revenue Protection:** $6,565,755 - $882,000 = **$5,683,755 (~$5.68 Million Projection)**

Sensitivity Matrix: Net Projected Revenue Saved (100k Customer Base)

Retention Success Rate	$10/mo Credit ($120/yr)	$15/mo Credit ($180/yr)	$20/mo Credit ($240/yr)
25% (Conservative)	$4,060,000	$3,745,000	$3,430,000
35% (Baseline Expected)	**$5,684,000**	**$5,243,000**	**$4,802,000**
50% (Optimistic Engagement)	$8,120,000	$7,490,000	$6,860,000

6. Production-Ready Deployment & Verification Evidence

To demonstrate complete production readiness beyond an experimental notebook, the project includes a production deployment bundle verified through automated test harnesses:

Production Component	File / Artifact	Specification & Operational Function	Verification Status
Serialized Model Pipeline	`production/churn_pipeline.joblib`	End-to-end preprocessing, scaling, alignment, and XGBoost inference	**Verified (Loads in 2ms)**

Production Component	File / Artifact	Specification & Operational Function	Verification Status
CLI Scoring Engine	`production/predict.py`	Batch scoring of CSV/JSON files with configurable threshold	**Verified (CLI Functional)**
REST API Microservice	`production/app.py`	Flask/Gunicorn service with `/health`, `/predict`, `/batch_predict`	**Verified (HTTP 200)**
Automated QA Suite	`production/test_production.py`	Integration tests checking deterministic bounds, latency & API	**4 of 4 Tests Passed**
Inference Latency SLA	`SLA Benchmark`	Average inference latency: **12.98 ms** (95th percentile: 13.59 ms)	**Passes (< 50ms SLA)**
Containerization	`production/Dockerfile`	Docker specification for multi-worker container deployment	**Production Ready**

Automated Test Suite Output:

```
[PASS] All 7 production artifacts verified successfully.
[PASS] High-Risk Customer correctly scored: Prob=77.39%, Tier=CRITICAL RISK
[PASS] Low-Risk Customer correctly scored: Prob=1.71%, Tier=LOW RISK
[PASS] Average Inference Latency: 12.98 ms (Production SLA < 50ms satisfied)
[PASS] /predict & /batch_predict API responses valid (HTTP 200)
ALL 4 PRODUCTION VERIFICATION TESTS PASSED (100% SUCCESS)
```

7. Actionable Business Retention Recommendations

1. Annual Contract Migration Campaign

Target: High-risk Month-to-Month customers in months 3–9.
Action: Offer a guaranteed **$10/month bill discount** or free streaming add-on conditioned on signing a 12-month agreement. Migrating customers to annual terms reduces churn risk by **73%**.

2. Fiber Optic "Value Assurance" & Quality Guarantee

Target: All Fiber Optic internet subscribers (41.9% current churn).
Action: Bundle complimentary **Premium Tech Support** and **Online Security** for the first 6 months. Run automated line health diagnostics to fix speed issues before customers call to cancel.

3. Auto-Pay Payment Channel Incentivization

Target: Customers paying via Electronic Check (45.3% current churn).
Action: Provide a **one-time $25 bill credit** for enrolling in Automated Bank Debit or Credit Card Auto-Pay, migrating them into a cohort with only 16.7% churn.

4. First 90-Day "New Subscriber Onboarding Shield"

Target: New subscribers in their first 90 days of tenure.
Action: Deploy automated 30, 60, and 90-day satisfaction check-ins and dedicated onboarding callbacks to eliminate early friction.
